

BHaM Database: Experimental Data Collection Notes Affecting Data Quality

This document contains notes from each experimental data collection session. The notes have been transcribed and edited for length and clarity. The notes are organized by participant (B001 thru B030) with notes from both the Elbow and Wrist Session and Hand Session provided.

General Notes:

- All collections were led by a single graduate student (M.D.) with varying assistance from undergraduate and graduate students.
- All fine wire EMG were placed by a single graduate student (M.D.) with ultrasound guidance. The ultrasound probe was positioned and held by the same trained individual (A.B.) for all sessions.
- The sub-max tasks and wrist tasks from the Elbow and Wrist Session may have large spikes at start of the first trial due to the need to reach a required threshold for the Biodex to begin countdown.

Notes by Participant:

1. B001
 - a. Participant did not appear to give true efforts during maximum voluntary contraction (MVC) or range of motion (RoM) tasks; as a result, this participant is not recommended for analysis.
 - b. Elbow and Wrist Session
 - i. Although the RoM_Wrist_Practice was performed in the standardized task order, the actual RoM_Wrist task was not performed until immediately before the RoM_Elbow_Task.
 - ii. RoM torque readings will have an additional torque spike every rep that was being added by the Biodex for an unknown reason.
 - c. Hand Session
 - i. The participant had unique anatomy that led to uncertainty regarding OPP and EPB insertion locations. There is lower confidence in OPP than EPB, as the ultrasound image guidance shifted during insertion.
 - ii. The APL insertion is likely incorrect. Based on observed muscle activity, the APL channel for this participant could represent FDP, FDS, or EDC (most likely EDC). The signal showed expected activity in Index Flexion/Extension, but no activity in Ab/Adduction tasks. The signal also indicated deactivation in Index Flexion MVCs and had clear activity in Lateral Pinch. Although these observations are likely due to electrode misplacement, the participant not giving full effort could also contribute to these findings.
2. B002
 - a. No Comments
3. B003
 - a. Elbow and Wrist Session
 - i. No Comments

- b. Hand Session
 - i. The original Index Flexion MVC got overwritten and was redone at the end of the session. However, the original Index Flexion MVC (which was not saved) was used to determine the sub-max target for the Index Flexion 50 MVC.
- 4. B004
 - a. Elbow and Wrist Session
 - i. No Comment
 - b. Hand Session
 - i. FDI signal lost due to battery failure of EMG sensor during the Lateral Pinch MVC. For this reason, FDI signal is missing from the following tasks: Lateral Pinch MVC/50MVC, Tip Pinch MVC/50MVC, and Index Flexion MVC/50MVC.
 - ii. OPP signal lost due to battery failure of EMG sensor during Tip Pinch 50MVC. For this reason, OPP signal is missing from the following tasks: Tip Pinch 50MVC and Index Flexion MVC/50MVC.
- 5. B005
 - a. Elbow and Wrist Session
 - i. No Comment
 - b. Hand Session
 - i. No data from Jar Twist tasks because the 1D force sensor was broken.
- 6. B006
 - a. Elbow and Wrist Session
 - i. No Comments
 - b. Hand Session
 - i. FPB wires pulled out after muscle-specific MVCs. So, FPB signals are available for muscle-specific MVCs, but no other tasks.
 - ii. FDI signal lost before muscle-specific MVCs due to battery failure of EMG sensor.
 - iii. Participant reported difficulties with Thumb Opposition Tip-to-Pinky-Base due to discomfort from fine wire insertions.
- 7. B007
 - a. Elbow and Wrist Session
 - i. Two RoM_Elbow trials were done. It is recommended to use the second trial as it was better.
 - b. Hand Session
 - i. The EMG sensor box for FPB fell off the forearm after the last trial of Tip Pinch MVC; the fine wires were not pulled out of the muscle during this event. The box fell after the last trial and was re-attached, so all data is usable and this event does not affect other tasks. However, a large signal spike is noticeable in the Tip Pinch MVC trial as a result.

8. B008
 - a. Elbow and Wrist Session
 - i. No Comments
 - b. Hand Session
 - i. ADD signal lost due to battery failure of EMG sensor during muscle-specific MVCs. So, ADD signal is missing for all tasks.
 - ii. OPP signal lost due to battery failure of EMG sensor during Index Flexion 50MVC. As a result, OPP signal is only available for Tip Pinch MVC/50MVC and Index Flexion MVC.
9. B009
 - a. No Comments
10. B010
 - a. Elbow and Wrist Session
 - i. No Comments
 - b. Hand Session
 - i. Marker (2Med) fell off during Index Flexion MVC but was replaced after the task.
11. B011
 - a. Elbow and Wrist Session
 - i. No Comments
 - b. Hand Session
 - i. EPB insertion was positioned extremely proximal and lateral. Based on later participants requiring similar placement, this likely represents an anatomical variation present in some participants.
12. B012
 - a. No Comments
13. B013
 - a. Elbow and Wrist Session
 - i. EMG was turned off during the first trial of Isometric Elbow Extension Practice.
 - ii. Participant extended elbow to straightened position during Isometric Elbow Extension tasks.
 - b. Hand Session
 - i. No data from Jar Twist tasks as the 1D force sensor was broken.
14. B014
 - a. Elbow and Wrist Session
 - i. No EMG data during Isometric Elbow Extension tasks as the Trigno Analog Out was turned off.
 1. EMG was turned on during third repetition of Isometric Elbow Extension 50MVC.

- ii. A Delsys Quatro battery died during the last task, Biodex_RoM_Elbow. So FCR, FCU, FDP, and FDS signals are not available for this task.
 - b. Hand Session
 - i. Participant only recruited OPP and FPB with direct isolation, such as during muscle-specific MVCs.
 - ii. No data from Jar Twist tasks as the 1D force sensor was broken.
- 15. B015
 - a. No Comments
- 16. B016
 - a. Elbow and Wrist Session
 - i. Participant did not follow directions well from Biodex prompts.
 - b. Hand Session
 - i. EPL wire movement led to noisy signals and gradual pull out of the fine wires from the EPL. No notes were taken on when this started to occur, as it was not noticed until near the end of the session.
- 17. B017
 - a. The participant did not follow directions; as a result, this participant is not recommended for analysis.
 - b. Elbow and Wrist Session
 - i. All Biodex data is questionable as participant kept finding different ways to 'enhance' strength even when instructed not to. For example, the participant tried using their knees to assist with tasks.
 - c. Hand Session
 - i. OPP wire came out at some point during the session.
 - ii. Participant found creative ways to cheat (e.g., used second hand for isometric tasks) and continued to do so when instructed not to.
- 18. B018
 - a. Elbow and Wrist Session
 - i. Issues with BIC sensor. Sensor readings seemed wrong during Isometric Elbow tasks as it was not firing logically. The EMG sensor box was replaced afterwards, and then experienced a battery failure during Isometric Wrist Extension 50MVC.
 - b. Hand Session
 - i. Participant only recruited EPB and ADD with direct isolation, such as during muscle-specific MVCs, but otherwise had low signals.
 - ii. No data from Jar Twist tasks as the 1D force sensor was broken.
- 19. B019
 - a. Elbow and Wrist Session
 - i. No Comments
 - b. Hand Session
 - i. Participant had trigger finger of index finger of tested arm.

- ii. Participant wore a jacket for partial cover as the experimental room was 65°F due to building HVAC issues.
- iii. No data from jar Twist tasks as the 1D force sensor was broken.

20. B020

- a. Elbow and Wrist Session
 - i. No Comments
- b. Hand Session
 - i. Data from the Preinsertion Lateral Pinch and Preinsertion Index Flexion forces were not saved. The missing data was not discovered until after the participant had left.

21. B021

- a. Elbow and Wrist Session
 - i. No Comments
- b. Hand Session
 - i. Participant did not seem to use EPL much. Signals were verified between tasks to ensure that EPL sensor was recording and the wires were in EPL.

22. B022

- a. Elbow and Wrist Session
 - i. No Comments
- b. Hand Session
 - i. Participant had textbook coactivation on/off timing patterns during Thumb Opposition Tip-to-Pinky-Base.

23. B023

- a. Elbow and Wrist Session
 - i. No Comment
- b. Hand Session
 - i. No data from Jar Twist tasks as the 1D force sensor was broken.

24. B024

- a. Elbow and Wrist Session
 - i. No Comments
- b. Hand Session
 - i. FPL was active during MVC tasks, but noticeably less active during range of motion tasks. The inactivity during range of motion was verified to be true by checking the sensor and performing extra muscle-specific MVCs to verify it was not a sensor or wire issue.

25. B025

- a. No Comments

26. B026

- a. Elbow and Wrist Session
 - i. No Comments
- b. Hand Session
 - i. EMG channels appeared noisy, but noise source was not identifiable and the noise can be filtered out.

27. B027

- a. Elbow and Wrist Session
 - i. ECU signal was very noisy.
- b. Hand Session
 - i. No data from Jar Twist tasks as the 1D force sensor was broken.

28. B028

- a. Elbow and Wrist Session
 - i. EMG was turned off during the first two repetitions of Elbow Extension MVC Practice.
- b. Hand Session
 - i. No Comments

29. B029

- a. Elbow and Wrist Session
 - i. A Delsys Quatro battery failed during the last task RoM_Elbow. So FCR, FCU, FDP, and FDS signals are not available for this task.
- b. Hand Session
 - i. No Comments

30. B030

- a. Elbow and Wrist Session
 - i. Participant reported wrist fusion during data collection.
 - ii. Participant could not do RoM_Wrist task due to wrist fusion. Participant could not do RoM_Elbow due to wrist fusion in combination with Biodex lever weight. As a result, the recorded data was taken with the participant performing the tasks to a metronome of 60 Hz and 30 Hz without the Biodex arm. This means that EMG data is available but no forces. The measured joint angle ranges were 80° for RoM_Wrist and 140° for RoM_Elbow; these ranges were measured manually with a goniometer.
- b. Hand Session
 - i. No Comment